InfinityGAN: Towards Infinite-Pixel Image Synthesis

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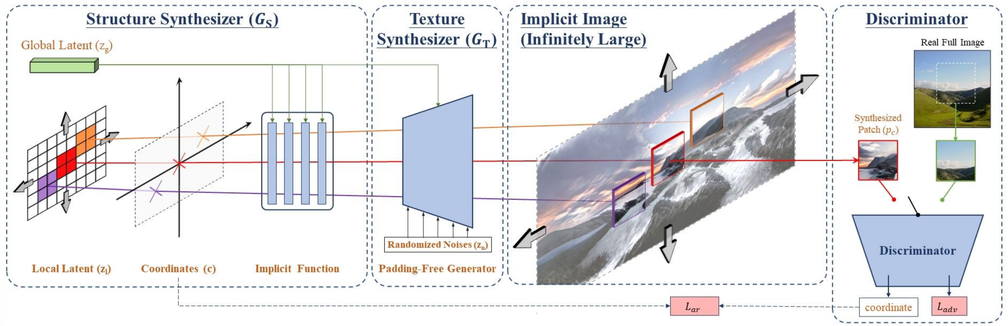
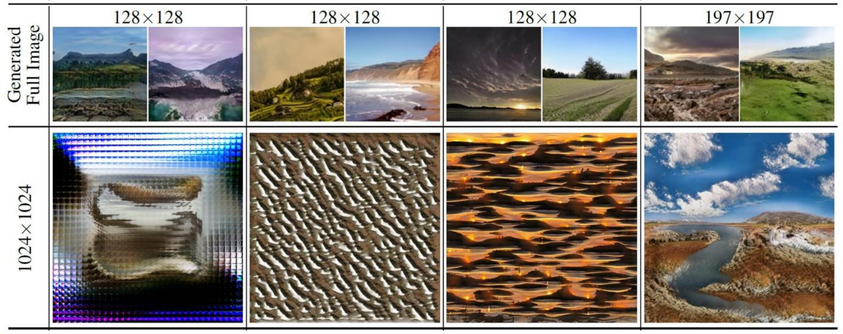
TL;DR We tackle the problem of synthesizing images at arbitrarily sizes, up to infinitely large.

Synthesizing *infinite*-pixel images from *finite*-sized training data.

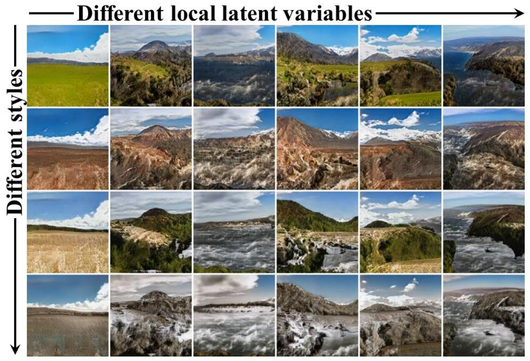
A 1024×2048 image composed of 242 patches, independently synthesized by InfinityGAN with spatial fusion of two styles. The generator is trained on 101×101 patches (e.g., marked in top-left) sampled from 197×197 real images. Note that training and inference (of any size) are performed on a ***single*** GTX TITAN X GPU.



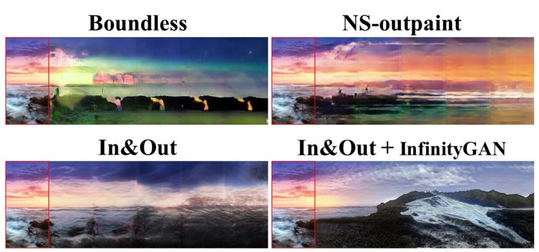
Without carefully taming the positional information in the generator, other frameworks cannot generalize to extended image sizes at testing.

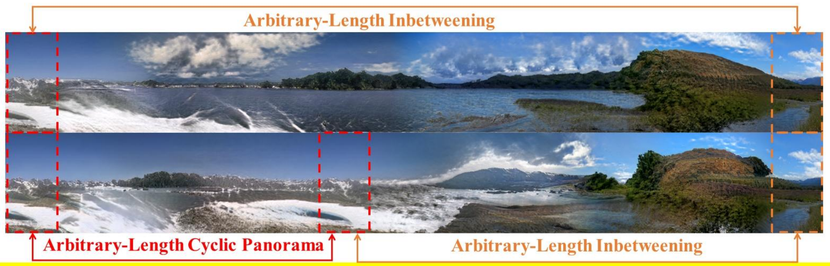


We show that the two sets of latent variables learn different and diverse appearances.



InfinityGAN + GAN-inversion (In&Out @ CVPR'22) = Seamless and arbitrarily-length image outpainting / inbetweening





More samples from models trained at higher-resolution settings.

A 256x10240 image

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Project Page

PyTorch